DEPARTMENT OF COMPUTER SCIENCE

Course Title: Ethical Hacking

Armitage - GUI Front-End for Metasploit

Objective

The objective of this experiment is to install and use Armitage, a graphical cyber-attack management front-end for the Metasploit Framework, in an isolated lab network. Students will learn to install Armitage and its dependencies, connect it to the Metasploit RPC service, organise targets using workspaces, discover hosts, identify applicable exploits, and carry out controlled exploitation and post-exploitation using its Meterpreter integration.

Procedure

1. Review Armitage and its role with the Metasploit Framework.
2. Install Armitage and the required Java, Metasploit and PostgreSQL components on Kali Linux.
3. Start Armitage, connect to the Metasploit RPC server and create a workspace.
4. Discover hosts and use "Find Attacks" to identify suitable exploit modules.
5. Run an exploit against the lab target and manage the resulting Meterpreter session.

Expected Result

After this experiment, you should be able to:

1. Explain what Armitage is and how it works with Metasploit.
2. Install Armitage and check that it is working on Kali Linux.
3. Start Armitage, connect to the RPC server and create a workspace.
4. Discover hosts, use "Find Attacks" and run an authorised exploit.
5. Manage a Meterpreter session and follow the lab safety rules.

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1. Introduction to Armitage

Armitage is a free, graphical cyber-attack management tool for the Metasploit Framework, originally created by Raphael Mudge. It connects to Metasploit's RPC service and visualises target hosts as icons on a canvas, recommends exploits based on the information gathered about each host, and exposes Metasploit's post-exploitation features (Meterpreter) through point-and-click menus instead of the msfconsole command line.

1.1. Architecture

Armitage itself does not run exploits directly. It is a Java client that talks to Metasploit's RPC daemon (msfrpcd), which in turn drives the Metasploit Framework's exploit, payload and auxiliary modules and stores host, service and session data in a PostgreSQL database. This separation is what allows several Armitage clients to connect to one shared Metasploit instance for team exercises.

1.2. Why It Matters

1. Armitage provides a graphical way to work with Metasploit instead of using only the command line.
2. "Find Attacks" compares discovered host and service information with available exploit modules.
3. Hail Mary can try applicable exploits automatically and can be used for demonstrations in a lab.
4. Multiple Armitage clients can work with the same Metasploit instance during a team exercise.
5. Meterpreter functions such as file browsing, screenshots, privilege escalation and pivoting can be accessed from the GUI.

2. System Requirements and Prerequisites

Armitage is a Java application, so it will run on Linux, Windows or macOS, but in a teaching lab it is almost always used on Kali Linux together with the Metasploit Framework that ships with it. Before installing, confirm the lab machine meets the following minimum requirements.

1. **Operating system**: Kali Linux 2023.x or later (or another Debian-based distribution with the Kali repositories available).
2. **RAM:** At least 4 GB; 8 GB or more is recommended when Metasploit's database and several Meterpreter sessions are active at once.

3. **Disk space**: Roughly 2 GB free for Metasploit Framework, Armitage, Java and the PostgreSQL database.
4. **Java**: OpenJDK 11 or later (installed automatically as a dependency of the armitage package on Kali).
5. **Network access**: the lab virtual machine must be on the same isolated network as the authorised target machines (for example, a host-only or internal VirtualBox/VMware network).
6. **Privileges**: administrative (sudo/root) access on the attacking machine, to install packages and manage the PostgreSQL and Metasploit services.

This experiment assumes the target(s) are lab virtual machines that you own or are explicitly authorised to test, such as Metasploitable2/3 or a deliberately vulnerable Windows/Linux image set up for the course. Armitage and Metasploit must never be pointed at systems outside this authorised lab environment.

3. Installing Armitage

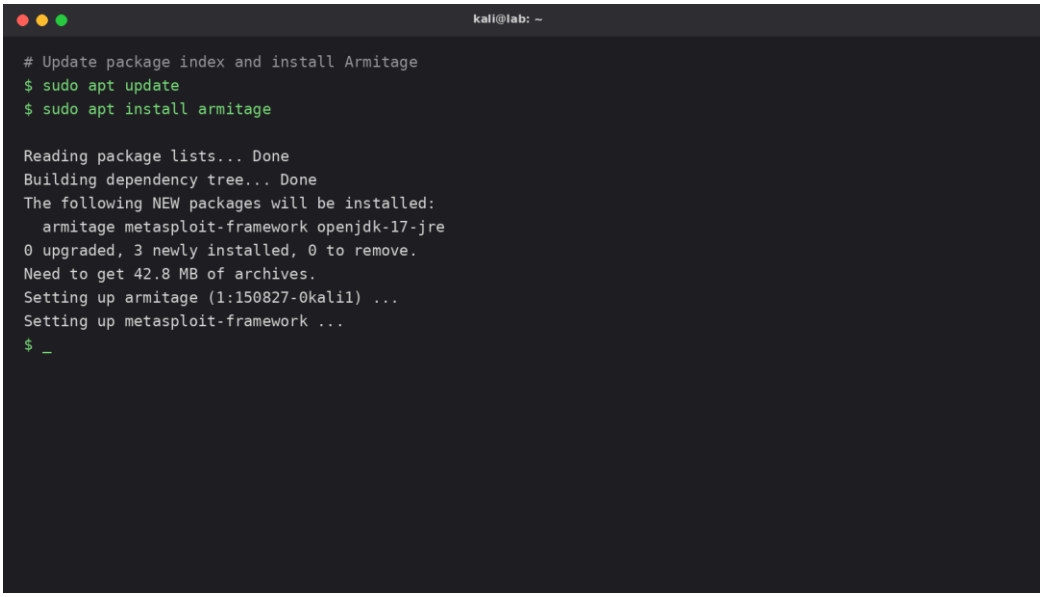
Step 1: Update the Package Index

Before installing new packages, refresh Kali's package lists so that the latest available version of Armitage and its dependencies are installed:

```
$ sudo apt update && sudo apt full-upgrade -y
```

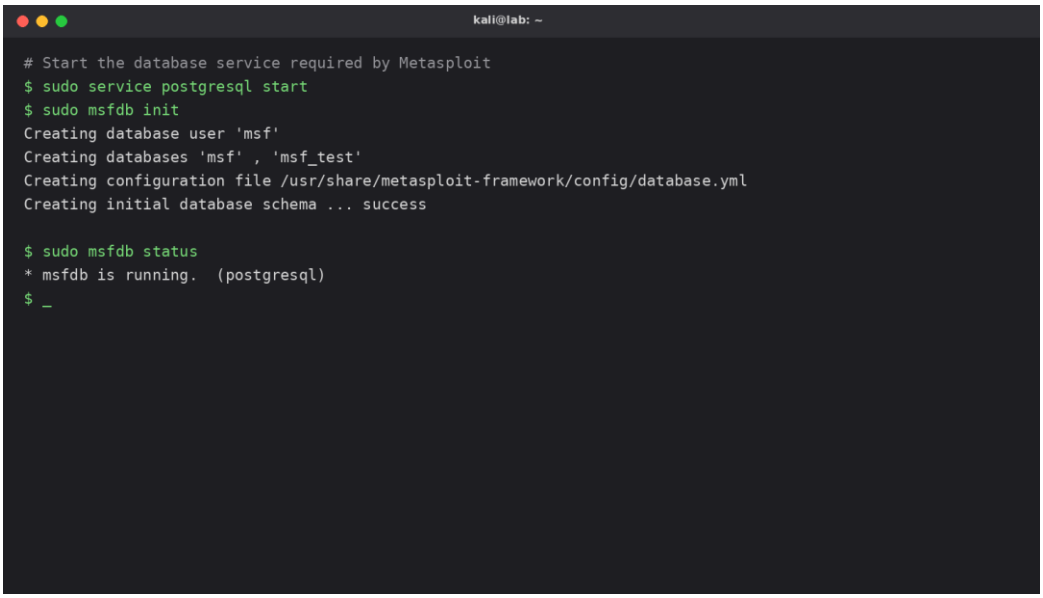
Step 2: Install Prerequisites and Armitage

On Kali Linux, Armitage is available directly from the standard repositories. Installing the armitage package automatically pulls in the Metasploit Framework and a compatible Java runtime as dependencies if they are not already present:



Step 3: Initialise the Database and Start Services

Metasploit (and therefore Armitage) stores host, service and session data in a PostgreSQL database. Start PostgreSQL and initialise the Metasploit database before the first run. This only needs to be done once per machine; afterwards, only the service needs to be started at the beginning of each lab session:



Step 4: Verify the Installation

Confirm that the individual pieces are in place before launching the GUI. "armitage --help" should print its command-line usage, and "msfconsole -v" should print the installed

Metasploit Framework version. If either command is not found, re-run Step 2 and check the terminal output for package errors.

1.$ armitage --help (prints Armitage's usage/version banner)

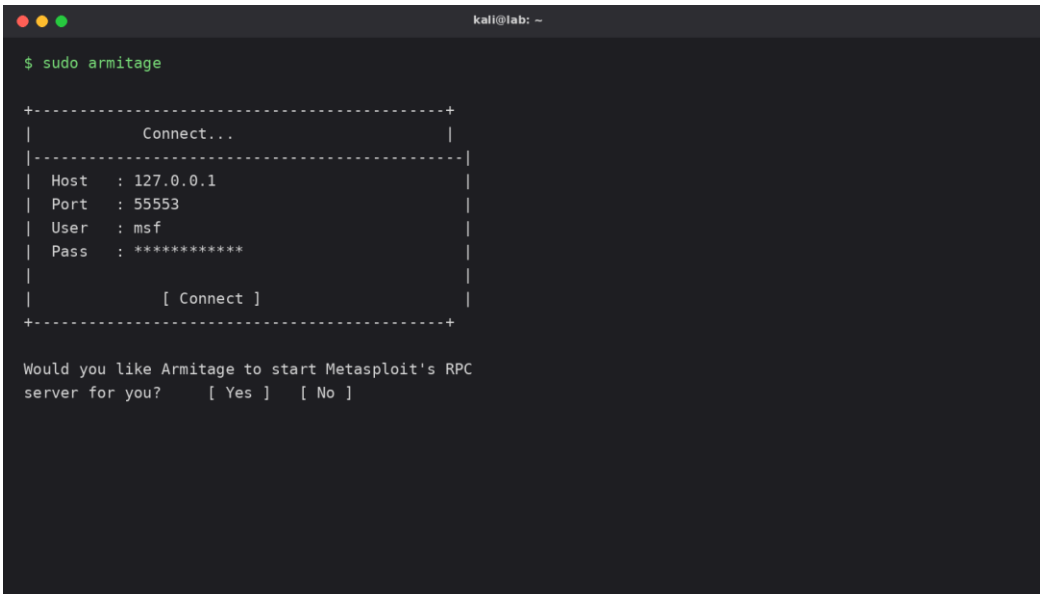
2.$ msfconsole -v (prints the installed Metasploit Framework version)

3.$ sudo msfdb status (confirms the database service is running)

4. Launching Armitage and Connecting to Metasploit

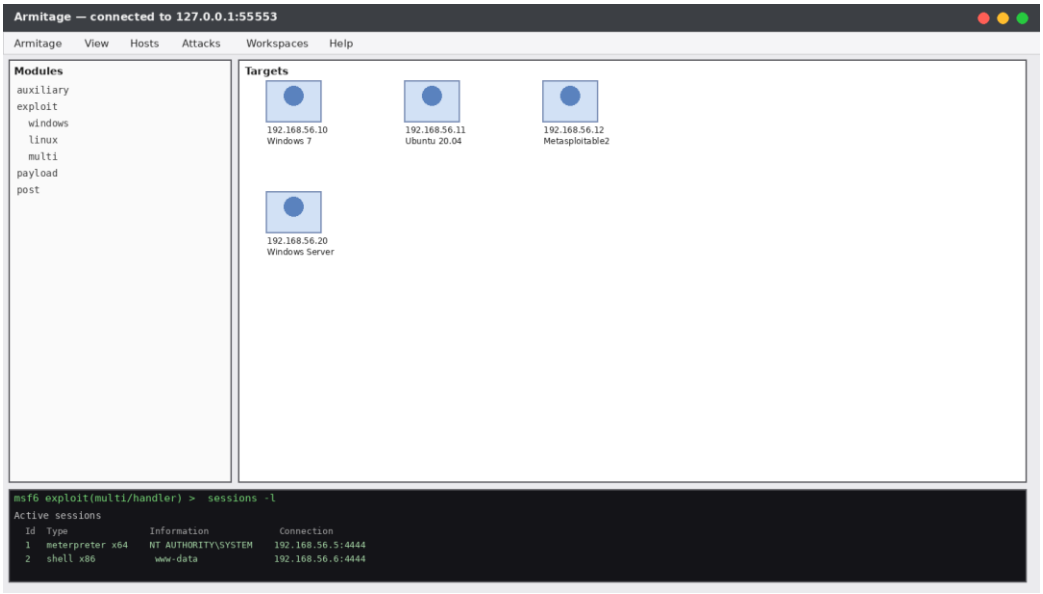
Step 1: Start Armitage

Launch Armitage with root privileges so that it can start the Metasploit RPC daemon if needed. It will prompt you to connect to Metasploit's RPC service on 127.0.0.1:55553; accept the default host, port and username, and allow Armitage to start the RPC server (msfrpcd) automatically if it is not already running. The very first connection can take a minute or two while Metasploit builds its module cache.



Step 2: Explore the Main Interface

Once connected, Armitage's main window shows three areas: a Modules panel on the left listing Metasploit's auxiliary, exploit, payload and post modules, searchable by keyword; a Targets panel showing every host added to the current workspace as an icon, colour-coded by operating system and exploitation status; and a console area at the bottom with a tab for each active shell or Meterpreter session, plus a raw msfconsole tab for typing commands directly when needed.



Step 3: Understand the Menu Bar

1. Armitage menu - preferences, team server settings and exit.
2. View menu - show or hide the Modules, Console and Targets panels and change the host view.
3. Hosts menu - add hosts, import a host list, or run Nmap/MSF scans.
4. Attacks menu - build the attack database and run "Find Attacks".
5. Workspaces menu - create, rename or switch between workspaces.

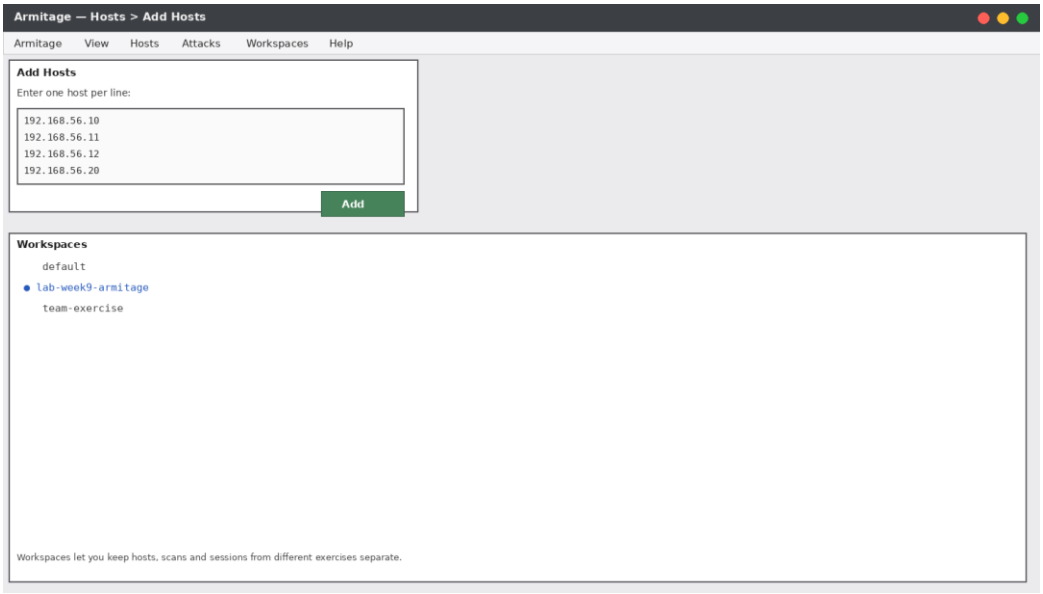
5. Workspaces and Host Management

Step 1: Create a Workspace for This Exercise

Workspaces > Manage lets you create a new, empty workspace so this lab's hosts, scans and sessions stay separate from any other exercise sharing the same Metasploit database. Name it something identifiable, such as "lab-week9-armitage", and switch to it before adding any hosts.

Step 2: Add Hosts Manually or by Import

If you already know the IP addresses of the authorised lab targets, use Hosts > Add Hosts and enter one address per line. Alternatively, Hosts > Import Hosts accepts a text file, or output from a previous Nmap/Nessus scan, which is useful when a target list has been prepared in advance.



Step 3: Discover Hosts and Services with Nmap

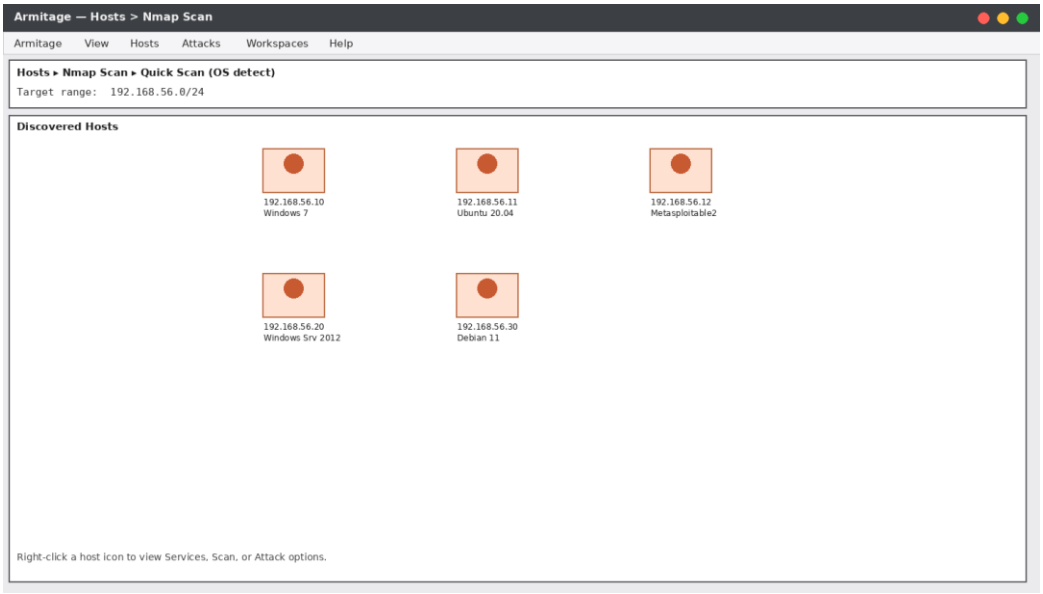
When the target range is not already known, use Hosts > Nmap Scan and choose a scan profile such as Quick Scan (OS detect) or Intense Scan for the authorised lab network range (for example, 192.168.56.0/24). Armitage runs Nmap in the background and populates the Targets panel with every host it finds, together with any operating system it could fingerprint.

Double-clicking a host, or right-clicking it and choosing Services, lists the open ports and service banners Armitage has recorded for that host. This service information is what "Find Attacks" (Section 6) uses to narrow down which exploit modules are worth trying.

6. Scanning and Attacking a Target

Step 1: Discover Hosts with Nmap

(If not already done in Section 5) Use Hosts > Nmap Scan > Quick Scan (OS detect) and enter the authorised lab network range. Discovered hosts appear as icons in the Targets panel, labelled with their IP address and, where identifiable, operating system.

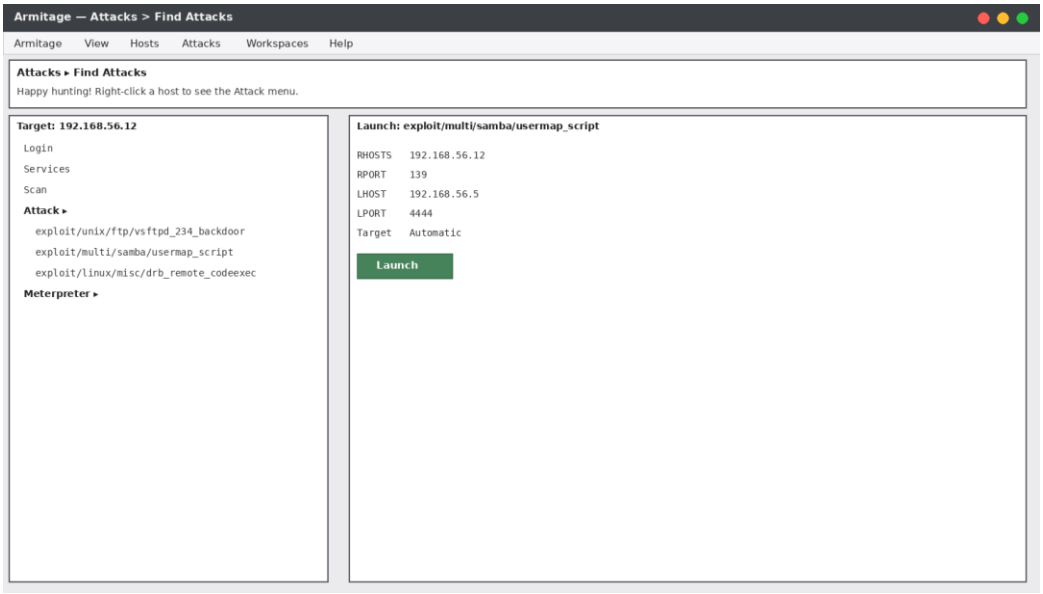


Step 2: Find Attacks

Select one or more hosts and use Attacks > Find Attacks so Armitage compares each host's recorded services against its exploit database and tags every host with an Attack submenu of matching modules. This step can take a little while the first time it builds the attack database, since it must load Metasploit's full module list.

Step 3: Launch an Exploit Manually

Right-click a target, choose Attack, and pick a module from the list Armitage suggested. Review or adjust its options - RHOSTS (target), RPORT (target port), LHOST/LPORT (your listener address and port for a reverse payload), and Target (the specific OS/version variant the module should use) - then click Launch to run it against the authorised lab target.



Step 4: Try Hail Mary for Automated Exploitation (Optional)

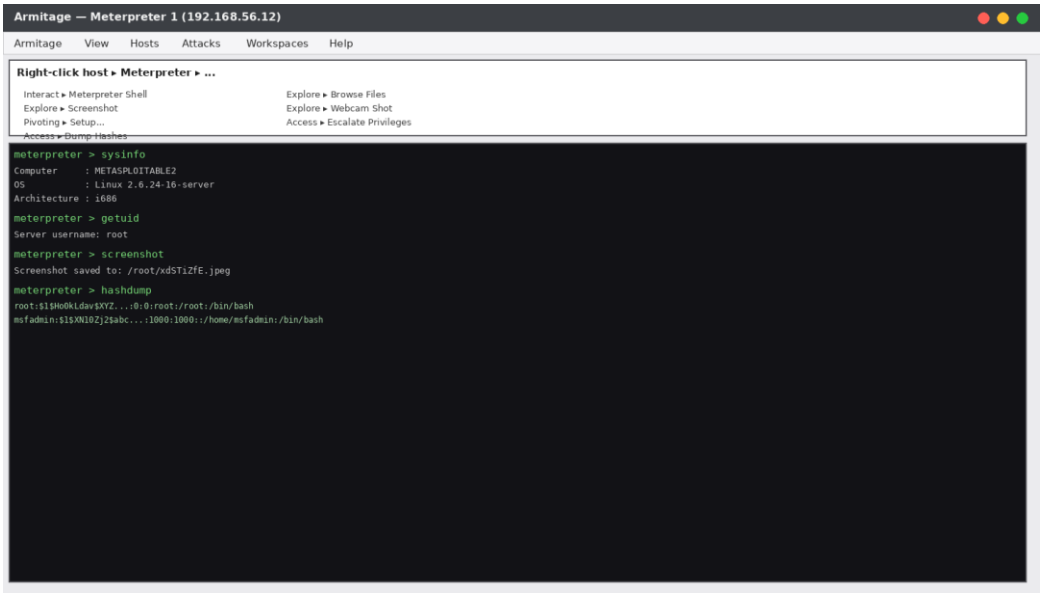
Attacks > Hail Mary runs every exploit Armitage judges relevant against all hosts in the current workspace, one after another, and reports which attempts succeeded. It is noisy and not selective, so it is best used only as a quick demonstration in an isolated lab, never as a substitute for the more targeted, service-by-service approach in Step 3.

7. Post-Exploitation with Meterpreter

A successfully exploited host turns red in the Targets panel and gains lightning-bolt icons, indicating one or more open sessions. Each active session also opens as its own tab in the console area at the bottom of the window, so multiple compromised hosts can be managed side by side.

Step 1: Interact with the Session

Right-clicking a compromised host and choosing Meterpreter > Interact opens a command tab where Meterpreter commands (sysinfo, getuid, ps, and so on) can be typed directly, in addition to the point-and-click actions described below.



Step 2: Explore the Target

1. Explore > Browse Files - opens a file browser for the lab target and supports file transfer.
2. Explore > Screenshot - captures the current desktop of a supported Windows lab target.
3. Explore > Webcam Shot / Show Desktop - additional capture options when supported by the lab target.

Step 3: Escalate Privileges and Pivot

1. Access > Escalate Privileges - runs suitable local privilege-escalation modules for the target OS.
2. Access > Dump Hashes - retrieves local password hashes for offline analysis on an authorised lab target.
3. Pivoting > Setup - routes Metasploit traffic through the compromised lab host to another network segment.

8. Troubleshooting and Good Practice

8.1. Common Issues

Symptom	Likely Cause / Fix
Armitage cannot connect to 127.0.0.1:55553	Run "sudo msfdb status" and "sudo service postgresql start"; retry the connection and allow Armitage to start msfrpcd.
"Connection refused" immediately on launch	The RPC daemon has not finished starting; wait 30-60 seconds after the first launch and try Connect again.

Find Attacks takes a very long time	Normal on the first run per session - Armitage is indexing Metasploit's full module list; subsequent runs are faster.
Exploit fails with "no session created"	Check LHOST/LPORT match a reachable listener address, and that the target's firewall allows the chosen port.
GUI fonts/icons render incorrectly	Usually a Java/OpenJDK version mismatch; reinstall with "sudo apt install --reinstall armitage openjdk-17-jre".